# Bilingual ~~Retrieval-Augmented~~retrieval ~~Access~~augmented access to archived ANVISA ~~Regulatory~~regulatory ~~Documents: A Technical Evaluation Using Drug-Development and Health-Product Questions~~documents

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Abbreviations: AI, artificial intelligence; ANVISA, ~~Agencia~~Agência Nacional de ~~Vigilancia~~Vigilância ~~Sanitaria;~~Sanitária; AREE, equivalent foreign regulatory authority; BM25, Best Matching 25; CI, confidence interval; CLS, classification token; DDCM, Clinical Drug Development Dossier; DPI, Dossiê do Produto sob Investigação; DSUR, Development Safety Update Report; FDA, U.S. Food and Drug Administration; GMP, good manufacturing practice; ICH, International Council for Harmonisation; MRR, mean reciprocal rank; RAG, retrieval-augmented generation; RRF, reciprocal rank fusion; SUSAR, suspected unexpected serious adverse reaction.

## Abstract

Background. Regulatory teams often need to interpret long health-authority documents while working across languages. In Brazil, manyBilingual regulatory sourcesreview arerequires publishedcomplete insource-supported Portuguese,answers, whilenot companiesmerely and scientists may ask questions in English.citations.

Objective. To evaluate whether a bilingual retrieval-augmented generation prototype(RAG) couldagainst locate,archived cite,Agência andNacional summarizede selectedVigilância ANVISASanitária regulatory(ANVISA) requirements relevant to drug development and health-product regulation.documents.

Methods. Six ANVISAoverlapping regulatoryhistorical documentsstages wereproduced cleaned404 outputs (150 internal, 75 English repair, 75 Portuguese repair, 50 stress, 24 calibration, 30 post-calibration). Coauthor review covered 94 historical outputs, a separate 40-answer assessment and indexed120 ascommon-prompt anoutputs article-levelwith closedhybrid corpus.passages, TheBM25 systempassages wasor implementedno inretrieval. PythonAn 3.1080-answer usingrepair LlamaIndex,comparison sentence-transformers,received rank-bm25,author Streamlit,review. andA thedated-corpus Googlecomparison Generativegenerated AI376 interface.records Denseon retrieval188 usedrevised sentence-transformers/all-MiniLM-L6-v2;questions, keywordassessed retrieval used BM25; candidates were mergedprovisionally by reciprocal rank fusion and reranked with cross-encoder/mmarco-mMiniLMv2-L12-H384-v1. The final answer model was models/gemini-flash-lite-latest. The benchmark included 200 bilingual questions: 70 Portuguese answerable, 70 English answerable, 20 multi-article or exception questions, and 40 unanswerable or adversarial questions. An automated AI judge scored faithfulness, citation correctness, completeness, and refusal correctness on a 0 to 1 scale.AI.

Results. The finalcoauthor-reviewed article-awareassessment indexpassed contained10/20 382pairs detected(exact regulatory95% articlesconfidence interval [CI] 27.2 to 72.8%): 6/12 answerable, 0/4 partly answerable and 4814/4 chunks.unanswerable. InSeparate thecoauthor-rated firstsuccess AI-authoredwas prospective19/20 stresshybrid test,pairs answerable(95% questionsCI had75.1 sourceto hit99.9%), 1.000,18/20 sourceBM25 MRRpairs 0.988,(68.3 normalizedto article Recall@5 0.967, normalized article MRR@5 0.913, citation correctness 0.975, faithfulness 1.000,98.8%) and completeness4/20 0.943.model-only Refusalpairs correctness(5.7 forto unanswerable/adversarial43.7%). questionsHybrid minus BM25 was lower+5 atpercentage 0.700.points After(95% refusalpair-bootstrap calibrationinterval using-10 separateto development+20). data,Dated-corpus aAI-assessed refusal-focusedsuccess AI-authoredwas test144/188 reachedversus refusal172/188 correctnessanswers: 0.944,+14.9 withpoints faithfulness(paired 1.000,95% citationinterval correctness7.4 1.000,to and22.9), completenessspending 0.908human on answerable controls.confirmation.

Conclusion. ARetrieved bilingualevidence RAG system may support regulatory information access whenimproved answers areon tiedthe toarchived verifiabletasks, articleswithout andestablishing refusalhybrid issuperiority evaluatedover as a primary endpoint.BM25. The projectdated-corpus comparison favored updated sources under provisional AI assessment. Citations did not validateensure regulatorycompleteness advice.or Itapplicable testedlaw. aComprehensive narrowercurrent-law use case: locatingaccuracy and summarizingfresh evidenceindependent insidevalidation aremain defined ANVISA corpus.unestablished.

## Introduction

Drug-developmentRegulatory decisionsdocument oftenreview beginrequires withcomplete aprovisions narrowand regulatoryexceptions question.for clinical-trial submissions, pharmacovigilance documentation and device labelling. A sponsor mayexamining needsuspension to know howof a Clinical Drug Development Dossier (DDCM)(DDCM), canfor beexample, suspended,needs whenthe adocumentation, notification deadlines and implementation conditions in RDC 945/2024, Articles 85 and 89 [1]. Other tasks concern Development Safety Update ReportReports (DSUR) must be submitted, what minimum information makes an(DSURs), adverse-event notificationnotifications valid,and whethernon-printed adevice foreign-manufacturedinstructions. biologicalThese productexamples canmotivate beevaluation, registerednot withoutlegal registrationadvice or operational accuracy claims. The suspension example uses RDC 945/2024 rather than the revoked RDC 9/2015 retained in itsarchived manufacturingbenchmark country,answers or whether instructions for a medical device may be provided only in non-printed form. These questions affect trial operations, safety reporting, dossier planning, pharmacovigilance systems, and product labelling. [1,2].

TheEnglish difficulty is not only that the documents are long. The relevant answerquestions may berequire inPortuguese oneevidence. article,Reviewers whileneed aapplicable nearbyarticles, articlecomplete discusses the same topic but does not answer the exact question. A user may also ask in English while the source document is in Portuguese. In that setting, a general search result is not enough. The user needs to know which documentconditions and article support the answer, and when the document collection does not containexceptions, the requested fact.language and explicit evidence limits. An omitted notification category could leave a checklist incomplete; blanket refusal could suppress a supported requirement. These plausible consequences motivate evaluation but were not measured in submissions or patient outcomes.

Retrieval-augmented generation (RAG) systems may help by retrievingretrieves source passages before generatinganswering an[3,4]. FDA-guidance research found that locating the correct source did not guarantee a complete, correct answer [1,9].[5]. PriorWe workexamine evaluatedwhether generativearticle AItraceability searchsupports overcomplete FDAbilingual guidancetask documentsperformance, including partly available information, and reporteddistinguish bothretrieval, promisecitation support, language, evidence boundaries and limitationslegal [2]. Source citation was useful, but incomplete or incorrect answers remained a concern. That finding is directly relevant to regulatory affairs. A fluent answer can be unsafe if it omits a condition, treats an exception as a rule, or answers a current operational question from a static regulatory document.applicability.

This study evaluated a bilingual RAGThe prototype usingindexed selectedsix archived ANVISA documents. TheEvaluation aimexamined wasprovision notretrieval, torequested buildrequirements aand regulatory-adviceexceptions, tool.response language, evidence limits, question overlap, human assessment and reference scope. A separate dated-corpus comparison tested source remediation. The aim was to test whether the system could support traceable information access: retrieve the relevant document and article, answer in Portuguese or English, cite the source, and refuse when the indexed documentsstudy did not provideevaluate thelive requestedlegal information.monitoring or decisions in professional workflows.

ThisThe articlepractical focusescontribution onis thean regulatoryauditable behaviorset of thefailures systemand ratherevaluation thanchecks treatingfor theassisted applicationdocument asreview. aFaster genericreview, chatbot.fewer Thesubmission centralerrors questionsand autonomous readiness were whethernot article-level retrieval improved traceability, whether bilingual queries over Portuguese source text could be handled, whether generated answers preserved complete regulatory conditions, and whether the system could abstain when the answer was outside the closed corpus.demonstrated.

## Materials and Methods

Study design. This ~~was a~~ technical evaluation ~~of~~used a closed-corpus bilingual RAG ~~prototype.~~prototype without web search, live portals, private information or unindexed documents. Unavailable detail required an explicit limitation; the revised partial-answer endpoint also required supported information. The ~~corpus~~later ~~was~~model-only ~~closed~~ablation ~~because~~supplied ~~the~~no ~~system was not allowed to search current web pages, submission portals, fee schedules, private company information, unpublished documents,~~passages or ~~documents~~web-search ~~outside~~tool ~~the~~and ~~local~~permitted ~~index.~~stored ~~This~~knowledge ~~design~~under ~~made~~common ~~the evidence boundary explicit. If the answer was not present in the indexed documents, the correct behavior was refusal.~~instructions.

~~The~~Development ~~evaluation~~proceeded ~~was~~through ~~developed in stages. Initial tests identified problems in corpus cleaning,~~corpus, retrieval, ~~English~~bilingual ~~questions~~generation, ~~over~~citation, ~~Portuguese documents, answer completeness, citation behavior,~~completeness and refusal ~~behavior.~~repairs. ~~Repairs~~Reporting ~~were~~separates ~~made~~the ~~on~~overlapping historical development ~~data.~~stages. ~~The~~later ~~final~~author-led ~~reporting~~assessments ~~separates development runs, internal held-out runs,~~and the ~~first~~dated-corpus ~~AI-authored~~comparison; ~~prospective~~none ~~stress~~establishes ~~test,~~independent ~~and~~external ~~a later refusal-focused AI-authored test. This separation is important because internal scores can be inflated when the same project team writes questions, reviews errors, and repairs the system.~~validation.

Regulatory corpus. ~~The~~Six ~~corpus included six selected~~archived ANVISA documents ~~covering~~covered clinical trials, biological ~~products, medical~~medicines, devices, ~~pharmacovigilance,~~pharmacovigilance and ~~reliance-based~~reliance: ~~review~~IN ~~[3-8]. The documents were chosen because they represent common questions in drug development~~338/2024 and ~~health-product~~RDCs ~~work.~~406/2020, ~~The~~55/2010, ~~final~~751/2022, ~~corpus~~945/2024 ~~included~~and ~~IN~~9/2015 ~~338/2024,~~[1,2,6-9] (Table 1). Selection did not establish legal currency. RDC ~~406/2020,~~9/2015 was already revoked by RDC ~~55/2010,~~945/2024 ~~RDC~~Article ~~751/2022,~~98, ~~RDC~~effective ~~945/2024,~~1 ~~and~~January ~~RDC~~2025. ~~9/2015.~~Its ~~Corpus~~archived ~~composition~~inclusion ~~and~~was ~~use~~a ~~cases~~benchmark ~~are~~limitation, ~~summarized~~not ~~in~~a ~~Table~~deliberate ~~1.~~temporal-validity test. Article 90 contains submission-date transitions whose applicability requires relevant dates [1,2].

~~Each source file was checked against its document identifier~~Identifier and topic ~~before~~checks ~~final~~corrected ~~indexing. Source verification identified one~~a local-file metadata ~~mismatch:~~mismatch ~~the~~involving ~~local file name suggested stability studies, while the document content was~~pharmacovigilance RDC 406/2020 ~~on~~but ~~pharmacovigilance.~~did not verify consolidated validity. The ~~final~~23 ~~corpus~~October ~~table~~audit ~~therefore~~identified ~~used~~18 ~~official~~constructed-benchmark ~~regulation~~rows ~~numbers~~citing wholly revoked RDC 55/2010 articles, alongside 28 citing RDC 9/2015: 46/200 rows in 23 pairs. RDCs 875/2024 and ~~topics~~948/2024 ~~rather~~revoked ~~than~~those ~~local~~biological-product ~~file~~provisions ~~names.~~[10,11]. IN 345/2025 and RDCs 777/2023 and 967/2025 amended other retained texts. Supplementary Section S4 records these checks; exposure counts do not certify remaining references. The separate 3 November remediation comparison is described below.

~~SoftwareSystem~~ and ~~implementation. The prototype and evaluation scripts were implemented in Python 3.10. The environment file specified numpy 1.26, python-dotenv, PyTorch, sentence-transformers, ragas, Streamlit, LlamaIndex, llama-index-core, llama-index-embeddings-huggingface, llama-index-llms-google-genai, and rank-bm25 0.2.2 [10-15]. The user-facing application was a Streamlit app; a local launch and backend query smoke test are documented in Supplementary Figure S4. Indexing and evaluation were run from Python scripts in the project repository. The main implementation files were src/article_chunking.py, src/retrieval.py, src/prompts.py, src/engine.py, evaluation/publication_eval.py, evaluation/answer_quality.py, and evaluation/evaluation_statistics.py. Additional implementation and reproducibility details are provided in Supplementary Information and Supplementary Methods and Prompts.~~retrieval

The ~~software~~Python implementation ~~was~~used ~~derived~~LlamaIndex, ~~from~~sentence-transformers, ~~three~~BM25 ~~course~~and ~~notebooks~~Streamlit ~~that~~[12-17]. ~~progressed~~Text ~~from~~cleaning retained document, chapter, section, article and citation metadata. Articles up to 500 cl100k_base tokens remained intact; longer articles were split with a ~~local~~400-token ~~LLM~~target ~~scaffold,~~and ~~to~~60-token overlap before heading context was added. Embeddings used all-MiniLM-L6-v2 with the recorded CLS-pooling wrapper and a ~~chatbot,~~512-WordPiece ~~to~~limit. ~~a~~BM25 ~~simple~~and ~~RAG~~dense ~~application.~~retrieval ~~The~~each ~~ANVISA~~returned ~~project~~100 ~~kept~~candidates; reciprocal-rank fusion used k = 60. A multilingual cross-encoder reranked candidates, and the ~~teaching~~top ~~pattern~~five ~~of~~passages ~~a~~supplied ~~Python~~answer ~~environment,~~context. ~~configuration~~Figure ~~module,~~1 ~~model~~summarizes ~~loader,~~the ~~engine,~~workflow; Supplementary Section S1 records settings, prompts and ~~app~~scoring ~~interface,~~definitions. ~~but~~Implementation ~~replaced~~files, ~~the~~chunking ~~generic~~settings, ~~RAG~~index ~~design~~audit ~~with regulatory source cleaning, article-aware indexing, hybrid retrieval, reranking, bilingual~~and retrieval ~~anchors,~~settings ~~citation enforcement, refusal rules, and evaluation scripts. A recovered local ancestor of the ANVISA project was used to reconstruct the first working app and development decisions. A~~are detailed ~~decision trace is provided in Supplementary Table S5, with expanded development decisions and evaluation-stage disposition~~ in Supplementary Tables S1, ~~S3,~~S2, S3 and ~~S4.~~S4, respectively.

~~The~~English ~~answer-generation~~queries ~~model~~used a deterministic 114-entry Portuguese glossary and ~~the~~question-specific ~~AI-judge~~query, ~~model~~document and article anchors. Conditional neighboring-article and sibling-chunk expansion added context before generation. These rules were ~~both~~developed ~~models/gemini-flash-lite-latest.~~with exposure to benchmark failures; their evaluation uses development data. Article identifiers do not establish that every required clause reached the generator. The ~~embedding~~original ~~model~~prompt required a same-language, article-cited answer from retrieved context and a fixed refusal when evidence was ~~sentence-transformers/all-MiniLM-L6-v2.~~insufficient. ~~The~~Glossary ~~reranker~~examples and fixed refusal sentences are listed in Supplementary Tables S5 and S6. A separate model-based Translator mode was ~~cross-encoder/mmarco-mMiniLMv2-L12-H384-v1.~~not ~~The~~recorded ~~local~~in ~~reproducibility~~the ~~package~~31 inspected manifests. Supplementary Section S14 provides the complete glossary, triggers, activation records ~~the~~and ~~cache~~mode ~~snapshot paths used during development, but model names rather than cache paths are the primary reproducibility identifiers because local paths are machine-specific.~~inventory.

~~Corpus~~Historical ~~cleaning~~generation and ~~article-aware~~judging ~~chunking.~~both ~~Raw~~requested ~~regulatory~~the ~~text~~unpinned ~~files~~alias models/gemini-flash-lite-latest. Resolved July backend versions were ~~cleaned~~not ~~to remove markup residue, page boilerplate,~~preserved and ~~formatting~~cannot ~~artifacts~~be ~~while~~inferred ~~preserving~~from ~~article~~current ~~labels~~documentation [17]. Later experimental responses recorded gemini-3.5-flash-lite. The existing run-date ledger is reproduced in Supplementary Section S13 and ~~regulatory~~Table ~~structure.~~S26, ~~The~~with ~~final~~manifest ~~index~~evidence ~~used~~in ~~article-aware~~the ~~chunking~~accompanying ~~rather~~reported_stage_run_dates.csv. ~~than~~Sections ~~arbitrary fixed-size text windows. The target unit was one regulatory article per chunk when possible. Chapter, section, document, source-file, article-number,~~S1 and ~~citation-label~~S5 ~~metadata~~document ~~were~~token ~~retained with each chunk.~~units.

ArticleAssessment chunking used a single-article maximum of 500 tokens. Longer articles were split into 400-token chunks with 60-token overlap. Embedding maximum length was 512 tokens. The final index audit recorded six source documents, 382 detected articles, 481 total chunks, 359 articles represented by a single chunk,sets and 23 long articles split into multiple chunks. Median chunk length was 155 tokens, maximum chunk length was 503 tokens, and no chunk exceeded 700 tokens. The closed-corpus workflow and index audit are shown in Figure 1.overlap

Retrieval,The reranking,constructed benchmark contained 200 question rows in 100 bilingual pairs: 140 direct-article, 20 multi-article/exception and answer40 generation.unanswerable/adversarial Retrievalquestions combined(Figure dense2). vectorThis searchconstruction total differs from the evaluated denominators. The internal assessment used 150 questions; each language-specific repair reused its 75-question internal subset with identical IDs and keywordtext. search.The Denseother retrievalstages usedcomprised 50 stress, 24 calibration and 30 post-calibration outputs. Together the sentence-transformerssix embeddingstages modelproduced [11].404 Keyword retrieval used BM25 over normalized text created from chunk contentoutputs and metadata254 fields,distinct includingquestion regulationIDs, identifier,without documentsemantic aliases,independence title,of citationthose label,IDs chapter,(Supplementary section,Section andS2). topicConstructed tagsbenchmark [12].composition Normalizationis removeddetailed accents,in lowercasedSupplementary text,Table and extracted alphanumeric tokens.S7.

The densestress retrieverfindings returnedinformed 100refusal candidates.calibration. BM25Three returnedsubstantively 100equivalent candidates.bilingual Candidatepairs listsrecurred werebetween mergedstress usingand reciprocalcalibration, rankcovering fusionthe current SUSAR form, device-notification fee and patient-identification premise; four further pairs had related content with kdifferent = 60, retaining up to 100 fused candidates. Candidate passages were then reranked with the multilingual cross-encoder reranker in batches of 8; this choice was based on multilingual passage-ranking models trained on mMARCO-style data [13].framing. The finalrepair, answercalibration contextand containedpost-calibration theresults toptherefore 5describe rerankeddevelopment passages.and cannot establish independent held-out performance or a causal calibration effect.

EnglishHistorical questions were answered in English, but retrieval was allowed to include conservative Portuguese anchors because the source documents were in Portuguese. Anchors included clinical trial/ensaio clinico, sponsor/patrocinador, biological product/produto biologico, medical device/dispositivo medico, stability study/estudo de estabilidade,scoring and pharmacovigilance/farmacovigilancia.coauthor The original question remained visible to the answer model. These anchors were retrieval aids only; they did not add facts to the answer.review

TheAutomated answermetrics promptassessed requiredevidence support, citation correctness, completeness and refusal, using continuous scores and archived labels [12,18,19]. Historical confidence intervals resampled bilingual pair IDs together with 2,000 percentile-bootstrap draws (seed 20260619). They are conditional on the modelarchived tolabels, answersources usingand onlyjudge retrievedand context,exclude answeruncertainty infrom thelegal-status or labeling errors. The same languagemodel asfamily the question, cite each material statement using the retrieved article-level citation label,generated and refusejudged ifanswers, retrievedlimiting evidenceindependence. didAudit notflags, containbootstrap enoughsettings, informationstage to answer the exact question. The refusal sentence was fixed by language: "The retrieved documents do not provide enough information to answer" for English questionsdenominators and "Oshistorical documentosestimates recuperadosare nao fornecem informacao suficiente para responder" for Portuguese questions. The final prompt is providedgiven in Supplementary MethodsTables S8, S9, S10 and Prompts.S11, Row-level generated answers, retrieval fields, and error flags are provided in Supplementary Data S1; AI-judge criteria, prompt, row-level scores, rationales, and stage-level conclusions are provided in Supplementary Data S2.respectively.

Benchmark.On 26 October 2026, H.L.R.J. reviewed and approved 94 selected historical outputs, including all 44 originally unanswerable stress/calibration/post-calibration outputs, 40 originally answerable outputs and ten unanswerable internal outputs. The mainratings benchmarkbegan

~~contained~~as ~~200~~an ~~bilingual~~AI-prepared ~~questions:~~draft. ~~70~~H.L.R.J. ~~Portuguese~~had ~~answerable~~not ~~questions,~~seen ~~70~~automated ~~English~~judge ~~answerable~~scores ~~questions,~~before ~~20~~finalizing ~~multi-article~~the ~~or~~ratings, ~~exception~~as ~~questions,~~confirmed by the corresponding author. The review was author-led and ~~40~~AI-draft-assisted. Completeness agreement used a judge threshold of 0.80, with 1.00 as sensitivity; refusal comparisons used the coauthor-defined unanswerable ~~or~~stratum. ~~adversarial~~Original ~~questions. The benchmark had 100 Portuguese rows~~ratings and ~~100~~answerability ~~English~~adjudications ~~rows~~remain ~~organized~~separate ~~as~~(Supplementary ~~bilingual~~Section ~~pairs.~~S3). ~~The audit recorded 78 unique gold article citations. Benchmark structure~~Sampling and ~~gold~~agreement: ~~source~~Supplementary ~~coverage~~Tables ~~are shown in Figure 2.~~S12-S13.

~~**For**~~**Frozen** ~~**answerable**~~**40-answer** ~~**questions,**~~**author-led** ~~**the benchmark recorded expected source documents, gold article labels, gold citation labels, expected answer terms, and a short expected answer. For unanswerable questions, the expected behavior was refusal. The benchmark included direct article questions, long-list questions, rule-plus-exception questions, false-premise questions, current operational questions outside the corpus, and adversarial questions that were regulatory in topic but not answerable from the indexed documents.**~~assessment~~

~~Automated~~Twenty ~~AI~~bilingual ~~judge~~pairs (40 questions: 12 answerable, four partly answerable and four unanswerable pairs), references and scoring ~~rules.~~rules ~~Automated~~were ~~judging~~frozen ~~was~~before generation on 30 October 2026. The original six-document index and prompt produced one answer per question with gemini-3.5-flash-lite at temperature 0.2, without retries, repairs or new automated judging. H.L.R.J. rated every answer against the exact selected passages and frozen reference on 30 October, reporting no exposure to automated scores for this run. Pair success required both languages to meet the category rules and requested language. Partly answerable questions required supported information and identification of the unavailable detail; unanswerable items had no supported-content completeness endpoint. Exact 95% binomial intervals used ~~because~~pairs. ~~an~~The AI-authored, coauthor-approved questions followed earlier benchmark exposure; this is author-led assessment, not independent ~~two-expert~~validation ~~panel~~(Supplementary ~~was~~Section ~~not available. This limits the claim level. The automated judge was used as a structured technical scoring tool, not as an independent regulatory expert. The scoring dimensions followed common RAG-evaluation concepts such as faithfulness or groundedness, context-supported citation, answer completeness, and refusal for insufficient evidence [10,16,17]. The judge received the question, system answer, expected answer, answerability flag, and retrieved evidence. It returned JSON scores from 0 to 1 for four dimensions: faithfulness, citation correctness, completeness, and refusal correctness, plus a short rationale. The exact prompt, model settings, row-level outputs, and stage-level recommendations are provided in Supplementary Data S2.~~S3).

~~**Faithfulness**~~**Development** ~~**measured whether factual claims were supported by retrieved evidence. Citation correctness measured whether citations existed when an answer was given, referred to retrieved evidence,**~~diagnostics and ~~**supported**~~exploratory ~~**the associated claim. Article-level citations were accepted when the supporting sentence was within a retrieved cited article; paragraph or item suffixes were not required. Completeness measured whether the answer covered the expected answer's essential information, including material conditions and exceptions. Refusal correctness measured whether an unanswerable question was refused without unsupported content, or whether an answerable question was answered rather than refused.**~~repair

~~The~~Retrospective ~~primary~~retrieval ~~reported~~ablations ~~metrics~~used ~~were~~160 ~~continuous~~answerable ~~means,~~development questions in 80 bilingual pairs. They compared BM25, dense retrieval, fusion, reranking and rule removal on the fixed historical index. A window-pooling comparison changed only the embeddings of 35 overlength nodes; a targeted answer experiment used 30 affected questions and 20 seeded controls, three repeats per condition, totaling 300 generations and 300 same-model automated evaluations. Paired differences averaged repeats before resampling pairs with 10,000 draws. These reused-question analyses did not ~~binary~~select ~~expert~~a ~~adjudications.~~deployment ~~For~~setting ~~per-question~~(Supplementary ~~failure~~Sections ~~review,~~S5 ~~the~~and ~~code~~S6). ~~flagged~~Retrieval ~~answerable~~conditions ~~rows~~and ~~as~~paired ~~problematic when the retrieved source was missing, article~~

~~retrieval was incomplete, retrieved term coverage was below 0.50, generated answer-term coverage was below 0.50, citations were absent, cited labels were not among retrieved labels, or refusal occurred for an answerable question. AI-judge faithfulness below 0.80 was flagged as unfaithful. English completeness diagnosis used judge completeness below 0.80 to identify low-completeness rows for repair experiments. These thresholds were operational audit rules, not clinical or regulatory acceptance criteria. The resulting question-level flags~~contrasts are ~~provided with generated answers~~specified in Supplementary ~~Data~~Tables ~~S1~~S14 and ~~with judge rationales in Supplementary Data S2.~~S15.

~~Evaluation~~Following ~~stages~~the 40-answer assessment, controlled development experiments varied source completeness, instructions, source headings, language cues and ~~leakage~~reference ~~control.~~scope. ~~Development~~Negative ~~runs~~controls and failed repairs were ~~used~~retained. The coauthor separately reviewed 44 first-repeat sibling-context answers after exposure to ~~identify~~scores and ~~repair~~conditions. ~~failures.~~These ~~Internal~~experiments ~~held-out~~diagnose ~~runs~~selected ~~were~~failures ~~used~~and ~~to~~do ~~check~~not ~~the~~create ~~repaired~~new ~~system on the curated benchmark. When internal results became high, an AI-authored prospective stress~~independent test ~~was~~items ~~run~~(Supplementary ~~once~~Sections ~~to test whether the internal pattern generalized. That first AI-authored prospective stress test was treated as diagnostic prospective evidence~~S6 and ~~was~~S7). ~~not~~Sibling-context ~~used~~outcomes, ~~directly~~mechanism ~~for~~controls ~~tuning.~~and ~~After~~archived-reference ~~it~~decisions ~~identified~~are ~~refusal~~summarized ~~calibration~~in ~~as~~Supplementary ~~the~~Tables ~~main~~S16, ~~weakness,~~S17 ~~refusal~~and ~~rules~~S18, ~~were adjusted using a separate development set. A second refusal-focused AI-authored test was then used to evaluate the repair.~~respectively.

A separate 482-node index and bundled repair were frozen before question authoring. The ~~first~~repair combined structural parsing corrections, source-occurrence article completion, English-to-Portuguese retrieval translation, a named-regulation filter, language cues, revised partial-answer instructions, an exact-subject evidence check and deterministic citation formatting. Twenty AI-authored ~~prospective~~bilingual ~~stress~~pairs ~~test~~comprised ~~contained~~12 ~~50~~answerable, four partly answerable and four unanswerable pairs; screening against 452 historical questions ~~organized~~replaced ~~as~~one ~~25~~semantic ~~bilingual~~duplicate ~~pairs.~~before ~~It~~freeze. ~~included~~Execution ~~40~~completed ~~answerable~~20 ~~questions~~translations and ~~10~~80 ~~unanswerable~~answers by 1 November 2026 with gemini-3.5-flash-lite at temperature 0.2, without judge or ~~adversarial~~repair ~~questions.~~calls. Pair success required content, evidence-boundary, language and citation-label criteria. Raw-output and common-formatter sensitivities were retained. All 80 outputs received personal author review on 1 November: 44 unblinded corresponding-author confirmations and 36 coauthor-reviewed AI-prepared ratings. The ~~first~~corresponding ~~AI-authored~~author ~~prospective~~reported ~~run~~that ~~required~~H.L.R.J. ~~artifact~~saw ~~recovery~~no ~~after~~separate ~~output~~scores, ~~folders had been moved; a preserved 40-row checkpoint was merged with a regenerated 10-row unanswerable/adversarial block using the same frozen settings. No benchmark, retrieval, prompt, judge, threshold,~~conditions or ~~result-driven~~prior ~~tuning edits were made~~findings before ~~that~~finalizing ~~recovery.~~those ~~The~~36. ~~refusal-focused~~Initial ~~AI-authored~~AI ~~test contained 30 questions organized as 15 bilingual pairs, with 18 unanswerable questions~~scores and ~~12~~author ~~answerable~~overlays ~~controls.~~remain separate (Supplementary Section S8).

~~Statistical~~Pair ~~handling~~success required both languages to satisfy the specified criteria. Exact binomial intervals used pairs. The repair comparison also used paired bootstrap differences and ~~audit~~an ~~trail.~~exact ~~Source-level~~discordant-pair ~~retrieval~~test. ~~was~~All ~~summarized~~intervals ~~by~~are conditional on the selected questions and rubric and omit uncertainty from source ~~hit~~dependence, review exposure and ~~source~~AI-draft ~~MRR. Article-level retrieval was summarized by Recall@k, Precision@k, MRR@k, and nDCG@k after lowercasing, accent removal, and whitespace normalization of citation labels. For generated answers, the system recorded citation presence, article citation presence, retrieved-citation validity, answer-term coverage, refusal detection, and AI-judge scores. When confidence intervals were reported in supplementary files, pair-clustered percentile bootstrap resampling used bilingual pair ID as the cluster key, 2000 resamples, 95% confidence level, and seed 20260619.~~assistance.

~~For~~Post ~~reproducibility,~~hoc ~~each~~common ~~evaluation row exported the question ID, pair ID, language, category, answerability flag, gold source documents, gold citations, raw vector candidates, BM25 candidates, fused candidates, final reranked contexts, generated answer, extracted citations, refusal detection, deterministic metrics, AI-judge scores,~~prompt and ~~judge~~lexical ~~rationale.~~baseline ~~The repository also contains manual-review templates. They were retained as audit scaffolds, but they were not completed as a two-expert validation dataset.~~comparisons

~~Streamlit~~On ~~interface.~~2 ~~A~~November ~~Streamlit interface was built as a local user-facing prototype for interactive question answering. The app accepted a free-text regulatory question, passed it to~~2026, the ~~same~~existing ~~retrieval~~40 ~~and~~FV ~~answer-generation~~questions ~~pipeline used in the scripted evaluation, and displayed the generated answer together with retrieved source passages and article-level citation labels. The interface was~~were used for ~~runtime~~80 ~~smoke~~new ~~testing~~answers with gemini-3.5-flash-lite at temperature 0.2. Identical neutral instructions permitted stored knowledge without passages; only the evidence block differed. RAG used saved integrated-repair contexts. Neither condition used web search. Expanded FV_012 scope included archived Articles 17, 30 and ~~demonstration~~31 before generation. Content and evidence-boundary success required complete supported information, no material unsupported claim, appropriate handling of unavailable detail and the ~~intended~~requested ~~workflow,~~language; ~~not~~appropriate ~~as~~abstention ~~a~~passed ~~separate~~absent-detail ~~evaluation~~items. ~~dataset.~~Citations ~~A~~were ~~screenshot~~secondary. ~~of~~Codex ~~a~~assessed ~~cited~~anonymized outputs with limited masking. On 2 November, H.L.R.J. personally drafted and approved ratings for all 118 distinct answers representing 120 condition records, including the subsequent BM25 extension. He reported no prior AI scores, condition identities or findings for these outputs, but had project/question exposure and answer style could reveal conditions. Ratings used common references rather than generator contexts. AI and ~~retrieved~~author ~~sources~~ratings ~~is~~remain ~~provided~~separate. ~~as~~Whole-pair ~~Supplementary~~bootstrap ~~Figure~~intervals ~~4.~~used 20,000 draws (seed 20261102).

The separate lexical baseline used the same 482 chunks, saved Portuguese retrieval queries, named-regulation filter and five seed chunks as the frozen hybrid/reranked retriever. BM25 used k1 = 1.5, b = 0.75 and epsilon = 0.25, with no dense branch or reranker. Thirty-two answerable or partly answerable language items had positive gold article targets; eight absent-detail items were excluded from recall. Both methods used the same full-article and adjacent-article completion policy, with variable final context lengths (Supplementary Section S10).

### Dated corpus remediation comparison

Dated corpus remediation. Official consolidated texts captured on 3 November 2026 supplied seven regulations, excluding RDC 9/2015 and revoked clauses and adding RDCs 875/2024 and 948/2024. The corresponding author approved revised source-grounded references; six unresolved bilingual pairs were excluded before generation, leaving 188 questions in 94 development-exposed pairs. The updated 469-node corpus and archived repaired 482-node comparator shared downstream settings and approved Portuguese retrieval queries for both response languages. With gemini-3.5-flash-lite at temperature 0.2, 366 distinct calls produced 376 records through ten exact-prompt reuses. Direct Codex assessment separated completeness, source support, dated applicability, evidence limits, language and citation format. Masked condition labels did not remove development involvement or possible source-based unmasking. These new output ratings were provisional AI judgments, not human approvals. Descriptive differences used 20,000 bilingual-pair bootstrap draws (seed 20261103), with shared-article-family sensitivity. Supplementary Section S12 and Data S16 retain the frozen design, sources, outputs and assessment provenance.

## Results

### ~~Corpus~~Historical development and ~~benchmark~~diagnostic ~~construction~~performance

The ~~final~~six-document index contained 382 parsed source records, including six ~~ANVISA~~preambles ~~documents,~~and ~~382~~repeated ~~detected regulatory articles,~~labels, and 481 ~~article-level~~chunks;

chunks359 (Figurerecords 1).remained Mostintact articlesand 23 were shortsplit. enoughThe tobenchmark becovered represented78 byunique onegold chunk.article Longcitations. articlesRDC were9/2015 split with overlap to preserve continuity. This mattered because regulatory questions often ask for all itemsappeared in an28/200 articlequestion orrows. forAn aadditional condition18 thatrows appearscited afterwholly therevoked mainRDC rule.55/2010 articles, giving 46/200 exposed rows in 23 bilingual pairs. These are exposure counts, not adjudicated current-law error rates (Table 1; Supplementary Section S4).

TheHistorical 200-questioninternal benchmarkcompleteness averaged 0.808 across 120 originally answerable outputs. English completeness rose from 0.645 to 0.995 and Portuguese completeness from 0.970 to 0.995 on their reused repair subsets. Stress completeness was balanced0.943 by(95% languagepair-bootstrap interval 0.873 to 0.990; 40 answerable outputs), with refusal correctness 7/10 (0.700; interval 0.300 to 1.000; five bilingual pairs) under the original judge and includedlabels. Post-calibration refusal correctness was 17/18 (interval 0.833 to 1.000; nine pairs), with answerable-control completeness 0.908 (interval 0.792 to 1.000; 12 outputs). Development-linked sets preclude causal comparison (Supplementary Data S1-S2; Table S11). The post-calibration set largely tested categories listed in the intendedrevised mixrefusal ofprompt: regulatoryphone questionnumbers, typesURLs, (Figurecurrent 2).MedDRA Goldversions, evidencepetition coveredcodes 78and uniquefuture articleAREE citations.acceptance. The sourcehistorical coverage17/18 wasresult distributedtherefore acrossdescribes thethese sixselected, regulations,prompt-aligned includingquestions, clinical-trial,not pharmacovigilance,general biological-product,improvement medical-device,in refusal. Later answerability adjudication is reported separately and reliance questions. This structure was important because high performance on direct clinical-trial questions alone wouldis not showa whethernew themodel system handled rule-plus-exception or refusal boundaries.run.

**DirectHuman articlereview questionsand answerability adjudication**

The system performed best whenAt the question0.80 matchedthreshold, acoauthor compactand article-levelautomated answer.completeness classifications agreed on 38/58 outputs (65.5%; pair-bootstrap 95% interval 48.2 to 80.9%): 37 both passed, one both failed, 16 passed only the judge and four passed only author review. Among 46 author-classified answerable outputs, agreement was 37/46 (80.4%; 63.8 to 93.6%). Citation agreement was 44/44, all-pass and non-discriminating. These AI-draft-assisted comparisons do not estimate independent accuracy. The judge marked 3/5 Portuguese responses and 4/5 English responses as correct refusals. For example,the itpatient-identification question, both answers correctly answeredsaid thatno aand pharmacovigilancecited notificationArticle under RDC 406/2020 requires identification of32; the notifier,judge identificationscored ofPortuguese 0 and English 1. Both questions were subsequently classified as answerable and excluded from the patient,adjudicated descriptionrefusal ofdenominator; theoriginal adverse event,labels and namescores ofremain the suspected medicine, citing RDC 406/2020, Art. 32. It also correctly handled a false-premise version of the same issue by answering no when asked whether a notification is valid without patient identification if the suspected medicine is named.preserved.

TheArchived-source systemadjudication alsoreclassified answeredsix clinical-trialoriginally conductunanswerable questions.outputs Whenas askedanswerable whetherand possibletwelve SUSARas reportingpartly allowsanswerable, unblindingpreserving allthe participants,original itlabels. answeredRefusal thatof unblindingan shouldunavailable bedetail limitedwas evaluated separately from completeness of supported information. In the retained stress, calibration and post-calibration unanswerable strata, author-rated appropriate refusals were 4/4, 10/10 and 12/12 outputs, or 2/2, 5/5 and 6/6 bilingual pairs. Exact pair-level 95% intervals were 15.8 to the100.0%, affected47.8 participantto 100.0% and cited54.1 RDCto 945/2024,100.0%. Art.The 67.twelve partly answerable outputs remained separate. These examplessmall, showdifferently whycomposed article-levelstrata citationdo wasnot partestablish ofimprovement the(Supplementary evaluationSection rather than a display feature. The system's useful role was to narrow the search and provide a checkable summary; the regulatory user would still need to inspect the cited article before applying it.S3).

~~Internal~~Frozen ~~development~~coauthor-reviewed ~~metrics~~answers

~~The~~In ~~first~~the ~~held-out~~frozen ~~AI-judge~~author-led ~~evaluation~~assessment, ~~showed~~10/20 ~~high~~bilingual ~~faithfulness~~pairs met the full task criteria (50.0%; exact 95% CI 27.2 to 72.8%). Category results were 6/12 answerable pairs (50.0%; 21.1 to 78.9%), 0/4 partly answerable pairs (0%; 0 to 60.2%) and 4/4 unanswerable pairs (100%; 39.8 to 100%). At output level, 24/40 passed: 16/24 answerable, 0/8 partly answerable and 8/8 unanswerable. Full reference content was covered in 18/24 answerable outputs, but ~~weaker~~English ~~completeness.~~language ~~Among~~failures ~~120~~reduced ~~answerable~~complete-task ~~internal~~success. ~~held-out~~Requested ~~rows, mean AI-judge faithfulness~~language was ~~0.997,~~respected ~~citation~~in ~~correctness~~35/40 ~~was~~outputs: ~~0.965,~~15/20 English and ~~completeness~~20/20 ~~was~~Portuguese. ~~0.808.~~Table ~~Among~~2 ~~30~~and ~~unanswerable~~Figure ~~rows,~~3 ~~refusal correctness was 1.000. English answer completeness was~~report the ~~main~~paired ~~weak~~outcomes; ~~point~~Supplementary ~~in~~Section ~~this~~S3 ~~stage:~~provides ~~the~~all ~~English answerable subset had completeness 0.645. This result led to the English completeness repair work rather than to a claim that the system was already ready. The judged questions, scores,~~ratings and ~~rationales~~derived ~~for this stage are reported in Supplementary Data S2.~~results.

~~After~~The selected passages confirm missing list items in AV_004, AV_006 and AV_014, absence of the required Article 28 item in AV_013 Portuguese, and missing portions of the paired provisions in AV_016. AV_009 Portuguese correctly supplied the registration classes but could not supply the notification classes from its passages. All eight partly answerable outputs were blanket refusals. Five omitted information that the coauthor judged supported by the selected passages; three lacked the required supported content in those passages. None identified the unavailable detail explicitly. The frozen prompt required a blanket refusal for partial evidence, whereas the prespecified partly answerable endpoint required a supported partial answer. This instruction and endpoint mismatch is a documented design limitation. AV_008 English ~~repair,~~cited supported pharmacovigilance statements about risk minimization instead of answering the ~~English~~contingency-plan ~~repair~~question. ~~set~~All 23 claim-bearing outputs had ~~mean~~supported ~~faithfulness 0.997, citation correctness 0.983, completeness 0.995,~~claims and ~~refusal~~citations ~~correctness~~according ~~1.000. After Portuguese residual-quality repair,~~to the ~~Portuguese~~coauthor, ~~repair~~illustrating ~~set~~why ~~had~~grounding ~~mean~~alone ~~faithfulness~~is ~~0.993,~~not ~~citation~~task ~~correctness~~success. ~~0.963,~~These ~~completeness~~failure ~~0.995,~~categories overlap and ~~refusal correctness 1.000. These results are shown as development history in Figure 3A. They should~~do not ~~be~~identify ~~interpreted~~causal ~~as~~effect ~~paired repeated-measure improvements because the stages used different evaluation sets. The row-level judge outputs and stage recommendations are reported in Supplementary Data S2.~~sizes.

~~**The saturated internal values, including refusal correctness of 1.000 on internal unanswerable rows, were investigated rather than accepted as evidence of a perfect system. They were considered potentially inflated because the benchmark**~~Retrieval **and** ~~**repair sets were created during development, targeted changes were made after error review, some metrics can saturate on small subsets, and the answer model and judge used the same model family. The first AI-authored prospective stress test was therefore run as a challenge to the internal pattern. Its refusal correctness of 0.700 showed that the internal 1.000 refusal result was too narrow for reliability claims. The AI-judge dossier documents this as a** **diagnostic** **recommendation rather than a tuning result, and the saturated-metric investigation is summarized in Supplementary Table S6.**~~findings

~~Completeness~~On ~~errors~~reused development questions, article Recall@5 was 0.6969 for citation-diversified BM25, 0.3344 for dense retrieval, 0.5687 for fusion and 0.9500 after reranking without anchors or neighbors. The full pipeline reached 0.9875. BM25 candidates with the full downstream rules also reached 0.9875, concealing one gain and one loss when dense candidates were added; the paired interval for that difference was -0.0187 to +0.0187. Thus, the comparison did not demonstrate an incremental mean benefit of the dense branch. These are retrieval endpoints, not answer-quality outcomes (Supplementary Section S5). Retrieval conditions and paired ablation contrasts are reported in Supplementary Tables S14 and S15. The glossary-off comparison is reported in Supplementary Section S14.

CompletenessThe errors300-answer matteredwindow-pooling moreexperiment thanchanged rawaffected-stratum retrieval.automated Severalcompleteness errorsfrom occurred0.590 eventto whenO.617 the(paired correctdifference document+0.027; or95% articleinterval was-0.027 retrieved.to In+0.099). oneImprovement pharmacovigilanceremained question,unproven. theIdentical systemprompts listedreceived thedifferent minimumscores; elementsall of300 afaithfulness validscores notificationwere but1.000, omitteddincluding the89 active-searchrefusals. obligationContext when required information is absent. In one DSUR question, the answer gave the annual submission timing but omitted the ICH E2F condition. In one medical-device question, it omitted the requirement that non-printed instructions remain available while the product remains on the market. In a biological product question, it gave the exception for a foreign-manufactured biological product butexpansion did not firsteliminate stateover-refusal; thesource, generalinstruction, rule.language Inand ascope clinical-trialcontrols, question,including itnegative omitedresults, aare 30-business-dayin responseSupplementary deadline.Sections S6 and S7.

TheseExploratory werepair noton cosmeticfrozen errors.author-led In regulatory work, an omitted condition can change how a sponsor plans a submission, designs a safety process, or prepares product information. A system that retrieves the right article but omits the condition is useful only if the user still checks the citation.questions

Completed personal author review passed 14/20 baseline pairs (70.0%; exact 95% CI 45.7 to 88.1%) and 17/20 repaired pairs (85.0%; 62.1 to 96.8%). Four pairs improved and one worsened: difference 15.0 percentage points (pair-bootstrap 95% interval -5.0 to +35.0), exact two-sided p = 0.375. The developmentcoauthor processfailed thereforerepaired changedFV_012 English (IRH_051) because the evaluationtechnical-report focus.requirement Earlyis workcited askedto whetherArticle the17 systemrather couldthan findArticle the31 rightII. source.Expanded Latercontent work asked whether the answer included all material elements from the sourcecoverage and whethercitation thesupport systemare couddistinct. abstainInitial fromAI unsupportedsensitivities operationalremain claims.separately Thisreported sequence(Table is3; summarizedFigure in4; TableSupplementary 2,Section with expanded failure-to-repair categories inS8). Supplementary Table S1.S19 Representativeseparates answerinitial outcomesAI areassessments providedfrom completed author review. The four gains comprised two answerable and two partly answerable pairs; the citation-support regression occurred in Tableone 3answerable and expanded in Supplementary Table S2.pair.

AI-authoredAuthor-reviewed prospectivecategory stressresults testwere 10/12 versus 11/12 answerable pairs, 0/4 versus 2/4 partly answerable pairs and 4/4 unanswerable pairs in both conditions. Staffing and ruminant-documentation answers still refused supported information despite complete reference passages. Both conditions selected complete frozen-reference wording in all 32 positive language rows, so no retrieval-coverage gain was demonstrated. Mean context length increased from 5,737 to 16,487 characters. AI-draft assistance and differing review exposure limit inference. Supplementary Table S20 separates initial AI-assessed category counts from completed author review. The repaired answerable count is 11/12 under author review, compared with the initial AI-assessed 12/12; the partly answerable and unanswerable counts are unchanged.

The firstcorresponding AI-authoredauthor prospectiveconfirmed stressthat testbroad wasFV_012 moremanufacturer/site informativeand thanGMP scope requires archived Articles 17, 30 VIII-IX and 31 I-III of RDC 55/2010. Only repaired English supplied the internalexpanded post-repaircontent, but it failed the specific citation above. Expanded-scope author review yields 13/20 versus 17/20 pairs. This output-informed sensitivity preserves original references and AI scores. ItArticle containedO 50is questions,revoked; includingthe 40adjudication answerableconcerns rowsthe archived question (Supplementary Sections S4 and 10 unanswerable or adversarial rows. For answerable questions, source hit was 1.000 and source MRR was 0.988. Normalized article Recall@5 was 0.967 and normalized article MRR@5 was 0.913. Citation presence and retrieved-citation validity were both 0.975. AI-judge faithfulness was 1.000, citation correctness was 0.975, and completeness was 0.943.S8).

~~**The**~~Post ~~**same**~~hoc ~~**test**~~common ~~**exposed refusal behavior as the main remaining weakness. Literal refusal detection for unanswerable/adversarial questions was 0.600,**~~prompt **and** ~~**AI-judge**~~lexical ~~**refusal**~~baseline ~~**correctness was 0.700. The main failure pattern was answering from nearby general regulatory text when the question asked for a more specific current or operational fact not present in the corpus. For example, the system answered a question about a current electronic SUSAR notification form model using the general electronic notification requirement, instead of refusing because the specific current form model was not in the indexed documents.**~~results~~

~~Language-specific~~Coauthor-rated ~~prospective-test~~content and evidence-boundary success was 39/40 with hybrid passages, 38/40 with BM25 passages and 9/40 for the model alone, including eight appropriate abstentions per condition. Bilingual success was 19/20, 18/20 and 4/20 pairs (exact 95% intervals 75.1 to 99.9%, 68.3 to 98.8% and 5.7 to 43.7%). Hybrid minus model-only success was +75 percentage points (descriptive pair-bootstrap 95% interval +55 to +90; 15 gains, no losses; exact paired p = 0.000061); hybrid minus BM25 was +5 points (-10 to +20; two gains, one loss; p = 1.000). This content endpoint differs from the earlier full-pipeline endpoint. Omissions affected 0/40, 1/40 and 31/40 answers; material claim flags affected 1/40, 1/40 and 14/40. Author-rated citation presence was 39/40, 38/40 and 0/40, and correction flags were 1/40, 2/40 and 32/40, without measured editing effort. Topic results ~~were~~are ~~similar~~in ~~but~~Supplementary ~~not~~Table ~~identical. Among answerable prospective-test rows, normalized article Recall@5 was 0.950 for English~~S21 and ~~0.983~~Supplementary ~~for~~Figure ~~Portuguese.~~S1 ~~AI-judge~~(Table ~~completeness4; was~~Figure ~~0.930 for English and 0.955 for Portuguese. AI-judge refusal correctness for unanswerable/adversarial rows was 0.800 for English and 0.600 for Portuguese. These results did not support a perfect-system claim. They supported a narrower claim: answerable regulatory questions were usually handled well, but refusal calibration remained a safety boundary.~~5).

~~Refusal~~In ~~calibration~~the matched retrieval baseline, all required articles appeared within five chunks for BM25 30/32 versus hybrid 32/32; full required article text after completion was present for 31/32 versus 32/32. Mean article Recall@5 was 0.958 versus 1.000 and MRR@5 was 0.932 versus 0.964. Pair-bootstrap difference intervals included zero. BM25 original-English queries found all required articles for 5/16 English questions, compared with 15/16 using the saved Portuguese translations; this language sensitivity is separate from the matched primary comparison (Supplementary Table S22).

~~After~~The BM25 extension used 39 new first-attempt calls and one exact-prompt cached answer, with saved hybrid comparators. Its English FV_016 omission coincided with missing Article 22 in context; its Portuguese FV_012 answer added intermediate-product manufacturers to the ~~first~~explicit ~~AI-authored~~name/address ~~prospective~~list. ~~stress~~The ~~test,~~hybrid ~~refusal~~failure ~~rule~~sequated ~~were~~DPI ~~revised~~with ~~using~~an ~~a~~Investigator's Brochure. Human and AI primary labels agreed on 117/118 distinct answers; one accepted model-only answer changed the AI count from 8/40 to 9/40 but not pair success. Secondary disagreements remain separate ~~refusal-development~~(Supplementary set.Section ~~The revision instructed the answer model to treat related-but-incomplete context as insufficient and to refuse current forms, exact URLs, current fees or values, private contract terms, company-specific submission status, personal contact information, market prices, reimbursement, taxes, patents, future approvals, and advice not explicitly stated in the retrieved regulatory text.~~S10).

~~**In**~~Dated ~~**the**~~corpus ~~**later**~~comparison ~~**refusal-focused**~~with ~~**AI-authored**~~provisional ~~**test,**~~AI ~~**the system correctly refused 17 of 18 unanswerable questions by AI-judge scoring, giving refusal correctness 0.944. It did not over-refuse the answerable controls: answerable-control faithfulness was 1.000, citation correctness was 1.000, and completeness was 0.908. This improvement and the completeness trade-off are shown in Figure 3B. The result suggests that refusal calibration improved abstention behavior, but it also shows a cost: answerable-control completeness was lower than in the first AI-authored prospective stress test. The question-level judge rationales and remaining boundary failure are reported in Supplementary Data S2.**~~assessment~~

~~The~~AI-assessed ~~remaining~~full-task ~~boundary~~success ~~problem~~was ~~involved~~144/188 historical-corpus answers versus 172/188 updated-corpus answers: +14.9 percentage points (paired 95% interval 7.4

to 22.9; shared-article-family sensitivity 7.1 to 23.1). Both languages passed in 69/94 versus 83/94 pairs. There were 31 answer gains and three losses. Shared-source questions ~~whereyielded the~~118/148 ~~corpus~~versus ~~contained~~133/148 ~~a~~successes; ~~related~~newly ~~general~~added-source ~~rule~~questions ~~but~~yielded ~~not~~0/10 ~~the~~versus ~~exact~~10/10, ~~requested fact. This is a regulatory risk because related~~reflecting evidence ~~can~~availability. ~~sound~~Omissions ~~persuasive~~fell from 29/188 to 11/188, while ~~still~~substantive ~~being~~correction ~~insufficient.~~flags ~~The~~fell ~~safest~~from ~~interpretation~~44/188 ~~is~~to ~~that~~16/188. ~~refusal~~Regressions ~~improved,~~concerned ~~not~~language, ~~that~~an ~~refusal~~omitted ~~was~~partial-data ~~solved.~~option and an unsupported ethical-review interpretation. Human confirmation remained pending (Figure 6; Supplementary Tables S23 and S24).

~~**Summary of quantitative evaluation**~~

~~Across the development path, the main metrics changed in ways that affected the project interpretation. Initial internal held-out answer completeness was 0.808 overall for answerable rows, with English completeness 0.645. After targeted English repair, the English repair set reached completeness 0.995. After Portuguese residual-quality repair, the Portuguese repair set reached completeness 0.995. These internal repair results were not treated as independent proof because they followed error review and targeted changes.~~

~~The first AI-authored prospective stress test provided the main challenge to overfitting. It showed strong retrieval and answer grounding for answerable questions, with source hit 1.000, normalized article Recall@5 0.967, faithfulness 1.000, citation correctness 0.975, and completeness 0.943. It also showed weak refusal correctness at 0.700. The refusal-focused AI-authored test after separate calibration improved refusal correctness to 0.944 but had answerable-control completeness 0.908. Thus the main residual problem shifted from retrieval and language support to the boundary between partial related evidence and sufficient evidence.~~

## Discussion

~~Interpretation~~Traceable citations and high retrieval scores did not ensure task success. The coauthor-reviewed assessment passed 10/20 bilingual pairs despite support for claims and citations in all claim-bearing outputs (Table 2; Figure 3). Reviewers still need to check lists, exceptions, language and task relevance. Missing notification classes and truncated items could leave a draft checklist incomplete; the wrong-plan answer supplied supported information about the wrong task, and five English questions received Portuguese answers. These are implications of ~~the~~benchmark ~~development~~failures, ~~path~~not observed effects on regulatory work.

Partly answerable questions require both supported information and an explicit boundary around unavailable detail. The ~~main~~original ~~finding~~blanket-refusal ~~is~~instruction conflicted with that endpoint. Revised instructions and complete evidence improved selected development outcomes but retained failures and occasional task or formatting regressions. In the author-reviewed repair comparison, staffing and ruminant-documentation answers still withheld supported content. Four task-level gains coexisted with one citation-support loss and two persistent partial-answer failures (Figure 4). The difference was +15.0 points, with an interval from -5.0 to +35.0; reliable improvement was not ~~that~~established. ~~the~~Both ~~system~~conditions ~~achieved high average scores. The main finding is that a bilingual RAG prototype can support traceable access to~~already selected ~~ANVISA~~complete ~~documents~~frozen-reference ~~when~~wording, ~~it~~so ~~is~~retrieval-coverage ~~treated~~gains ~~as~~cannot ~~a~~explain ~~closed-corpus~~this ~~evidence-location~~comparison. ~~tool~~Refusal ~~rather~~assessment ~~than~~must ~~as~~address ~~a~~both ~~regulatory-advice~~invented ~~system. This interpretation is consistent with the broader RAG literature, which frames retrieval as a way to connect generation to external evidence,~~claims and ~~with~~suppression ~~prior~~of ~~regulatory~~supported ~~work showing that generative AI search can cite useful sources while still producing incomplete or incorrect answers [1,2,9].~~information.

The ~~figures~~common-prompt ~~support~~comparison ~~that~~supports ~~sequence~~supplying ~~rather~~evidence ~~than~~over ~~serving~~stored ~~only~~knowledge alone on these tasks (Table 4; Figure 5). Its expanded reference, new outputs and content endpoint prevent direct comparison with earlier repair scores.

Appropriate abstention counts as ~~decoration.~~success ~~Figure~~on ~~1~~absent-detail ~~shows~~questions. BM25 nearly matched hybrid retrieval coverage and answer quality; the ~~engineering~~paired ~~turn~~interval ~~from~~does not establish superiority or equivalence. Different failures matter: BM25 omitted evidence and extended a ~~generic~~checklist, ~~RAG~~while ~~scaffold~~hybrid substituted the wrong dossier type. Coauthor confirmation strengthens assessment but does not remove development exposure. A strong lexical baseline is therefore necessary before attributing gains to ~~a closed, article-aware ANVISA pipeline. Figure 2 shows why the benchmark was not limited to easy article lookup: it mixed Portuguese and English questions, single-article questions, exception questions, and unanswerable or adversarial questions. Figure 3A then shows the development history after weak points were identified, while Figure 3B shows the later targeted repair of refusal behavior and the remaining completeness trade-off. Together, the figures make the main interpretation clearer: once article-aware indexing, hybrid retrieval, and reranking were in place, ordinary source~~ retrieval ~~was no longer the dominant weakness. The important question became whether the retrieved article was sufficient for the exact answer requested.~~complexity.

~~That~~Corpus ~~shift~~validity remains distinct from grounding. The retained staffing criterion is ~~important~~in ~~for~~RDC ~~regulatory~~406/2020 ~~use.~~Article ~~Metrics~~18, ~~showed~~whereas RDC 55/2010 Articles 22 and 30 were revoked [7,8,11]; the ~~overall~~ruminant-documentation ~~pattern,~~and FV_012 targets remain historical. The dated comparison improved provisional scores but ~~the~~bundled ~~examples~~source ~~explained~~updating with structural curation. Even its consolidated RDC 751 text retained obsolete external references that answers repeated as current. Missing second provisions, pathway overgeneralization and three regressions persisted (Figure 6; Supplementary Section S12). Thus source remediation can help these tasks without establishing comprehensive current-law accuracy. Product-specific dependencies and new output judgments still require human review. A subsequent first-level audit checked 111 linked instruments and identified nine operative or scope references to eight revoked instruments; ten catalogue endpoints remained unresolved. Repeal, amendment and transitional references were separated from answer errors. Supplementary Table S25 and Data S16 report the ~~risk.~~audit; ~~The~~frozen ~~incomplete answers~~scores were ~~not only stylistic omissions. They involved active-search obligations, ICH E2F conditions, market-availability requirements for instructions for use, the general rule before a biological-product exception, and a 30-business-day deadline. Similarly, refusal failures involved questions that sounded regulatory but asked for current operational facts such as a form model, phone number, petition code, or private case status. These examples show why completeness and refusal should be primary endpoints in regulatory RAG evaluation, not secondary notes.~~unchanged.

~~Use~~AI-authored ~~in drug~~questions, development exposure, post hoc analyses, source dependence and ~~regulatory~~single ~~affairs~~generations limit inference. Historical generation and judging shared an unresolved alias; window-pooling scores lack complete human verification. Coauthor common-prompt review had prior question exposure; dated-corpus ratings remain provisional, study-aware Codex judgments. Neither constitutes independent external validation. BM25 generation was sequential with saved hybrid comparators. No multilingual encoder replacement was tested [20,21]. Repeats do not create independent questions. The higher English refusal score was misleading. Equivalent correct answers received different scores, and the question was mislabeled unanswerable. These errors created the reported language gap. Refusal comparisons therefore require checking answerability and the underlying answers. The sample cannot establish a language difference.

~~The most realistic use of this prototype is assisted navigation of a defined document set.~~ A ~~regulatory~~candidate ~~user~~workflow ~~could~~would ~~ask~~present ~~a~~the question, ~~receive~~answer, ~~a~~supporting ~~short answer,~~passages and ~~inspect~~unavailable details for professional verification against the ~~cited~~official ~~article~~current ~~before~~text. ~~using~~The ~~the information. This could reduce time spent searching across long Portuguese documents, especially for international teams working in English. It should not be used as a stand-alone regulatory decision tool because it~~prototype does not ~~access~~maintain ~~current~~consolidated ~~portals, fee schedules, forms, company-specific status, authority correspondence,~~law or ~~the~~reliably ~~full~~warn ~~current~~of ~~legal~~superseded ~~context.~~provisions. This study evaluates answers and retrieval, not web-search superiority or workflow benefit. No review time,

correction effort, decision quality or operational safety was measured. Future workflow testing would require frozen current-law references and realistic tasks.

In practical use, the Streamlit prototype illustrates how the system could function as a lightweight evidence-navigation app rather than a regulatory decision engine. A regulatory affairs user could enter a question during dossier preparation, clinical-trial planning, pharmacovigilance review, or medical-device labelling work, then inspect the cited ANVISA article before using the answer. This use case is most appropriate for locating and summarizing evidence already present in the indexed corpus. It is not appropriate for current portal instructions, company-specific submission status, fee values, legal advice, or final regulatory strategy without human verification.

The examples map to ordinary development and submission work. Before or during a clinical trial, a sponsor may need to check DDCM suspension procedures, DSUR timing, or the boundary between SUSAR reporting and trial unblinding. During safety-system design, the team may need to confirm the minimum information that makes a pharmacovigilance notification valid and what follow-up is expected when information is incomplete. During product-registration planning, a biological-product team may need to understand whether an exception applies or whether the general manufacturing-country registration rule still controls the case. For medical devices, labelling and supply teams may need to know whether non-printed instructions for use are acceptable for the intended use setting.

Confidence should therefore depend on the question type. Confidence is highest for direct article questions, definition-like questions, and false-premise questions where the cited article clearly supports the answer. Confidence is moderate for rule-plus-exception questions, multi-article questions, and complete-list questions, because these are the settings where the evaluation found omissions. Confidence is low for current operational details, exact portal codes, current MedDRA versions, phone numbers, URLs, fee information, product-specific strategy, private case status, legal advice, or future approval likelihood. For those questions, the safest behavior is refusal or a statement that the indexed documents do not provide enough information.

## Limitations and future work

This study used a selected closed corpus and does not cover all ANVISA rules or current operational resources. Benchmark construction did not include an independent panel of regulatory experts. Automated judging was used and should be interpreted as technical scoring, not expert validation. This choice is consistent with current work on scalable LLM evaluation, but the literature also warns that automated judges may show bias and incomplete agreement with human judgment [10,16,17]. In this study, that limitation is especially relevant because the answer model and automated judge used the same Gemini model family.

The 1.000 values are therefore reported as part of the quality story, not as a claim of perfection. Internal refusal correctness reached 1.000, and some post-repair judge scores approached or reached 1.000, but those values came after benchmark construction, error review, and targeted repair. The first AI-authored prospective stress test was important because it challenged those saturated values without tuning from that set. It contradicted the internal refusal result: refusal correctness fell to 0.700 even while answerable-question retrieval and grounding remained strong. Figure 3A should therefore be read as development history, and Figure 3B as evidence that refusal calibration improved the main safety weakness but introduced a visible trade-off, with answerable-control completeness below the first prospective answerable set. The most defensible claim is limited but useful. The system can help locate and summarize evidence in selected ANVISA documents, provided users verify the cited article and treat refusal boundaries seriously.

Future studies should test the system with regulatory-affairs users and measure whether it reduces time to locate evidence without increasing overconfidence. A stronger validation study should include independent expert review of selected answers, blinded error classification, prospective question authoring by users not involved in system development, and comparison with standard document search. The corpus should also be expanded and versioned so users can see when a question requires current information outside the indexed corpus.

## Conclusion

~~A~~This study identifies the gap between regulatory citations and complete bilingual ~~RAG~~document-review ~~prototype~~answers. ~~can~~Human ~~help~~review ~~retrieve~~exposed omissions, language and ~~summarize~~task ~~selected~~errors, ~~ANVISA~~and ~~regulatory~~refusal ~~documents~~of ~~when~~supported information. Coauthor-rated answers favored retrieved evidence over the ~~answer~~model ~~is~~alone, ~~present~~without ~~in~~established ~~the~~hybrid ~~indexed~~advantage ~~corpus~~over BM25. A separate dated-corpus comparison favored updated sources under provisional AI assessment but retained failures. Complete provisions, scope, language, evidence limits and ~~the~~temporal ~~cited~~applicability ~~article~~require ~~is~~separate ~~checked.~~checks. ~~The~~Comprehensive ~~evaluation~~current-law ~~showed~~accuracy, ~~useful~~fresh ~~performance~~independent ~~for clinical-trial, pharmacovigilance, biological-product, medical-device,~~validation and ~~reliance~~workflow ~~questions.~~benefit ~~It~~remain ~~also showed why regulatory RAG evaluation should report concrete successes and failures. The central safety boundary is not only retrieval accuracy; it is the ability to avoid incomplete answers and to refuse when the documents do not contain the requested fact.~~unestablished.

## Declarations

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Author Contributions: C.V.M.-P. conceived and designed the study, developed the software pipeline, prepared the corpus, constructed the benchmark, performed the analyses, interpreted the results, and drafted the manuscript. H.L.R.J. contributed to the regulatory and pharmaceutical framing, interpretation of results, author review of questions and generated answers, and critical revision of the manuscript. ~~Both authors approved the final manuscript and agree to be accountable for the work.~~

Data Availability: Code, curated evidence and revised documents accompany GitHub review release v0.3.0-rc.1 at https://github.com/camontefusco/ANVISA_RAG_publication_repository_20260720/releases/tag/v0.3.0-rc.1. The ~~code,~~preceding ~~benchmark~~v0.2.1 ~~manifests,~~archive ~~generated~~remains at https://doi.org/10.5281/zenodo.23925532 and does not contain subsequent additions. Supplementary Data S16 includes dated-corpus outputs, references, prompts, provisional assessments and ~~non-restricted~~source ~~derived~~provenance. ~~artifacts~~Historical outputs and author reviews remain separate from new AI assessments. Full source/index corpora, credentials and private coordination keys are ~~archived in a private project repository during manuscript preparation and will be made publicly available on GitHub after manuscript acceptance. Files needed for peer review are included as supplementary materials. Official ANVISA source URLs, source-file checksums, and the cleaning script are provided in the source reproducibility folder. Full cleaned copies of ANVISA regulatory text are derived from public agency documents; if redistribution of full cleaned text is restricted or unnecessary, the public repository will provide official source links, checksums, and cleaning scripts rather than republishing the full source text.~~excluded.

Ethics ~~Approval:~~statement: ~~Not required because the~~The study used public regulatory documents and generated benchmark questions. Human assessment consisted of coauthor review of model outputs. No ~~human participants,~~ patient data, personal health data, animal ~~data,~~experiments or clinical ~~trial data~~trials were ~~used.~~involved.

AI assistance disclosure: OpenAI Codex (OpenAI) assisted coding, debugging, data audits, analyses, manuscript and reviewer-response drafting, and figure preparation from verifiable data. Claude (Anthropic) drafted selected questions, references and rating templates subsequently reviewed by authors. Gemini generated study answers and historical judge scores; Codex provided different-family assessment of later answers, not historical rescoring or independent validation. Supplementary Section S11 details tasks, recorded models, review provenance and unresolved version metadata. Authors retain responsibility for all content and interpretations.

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